Formula:

IC cost Calculation

(1) Cost per die [C/D] = $\frac{\text{Cost per wafer [C/W]}}{\text{Dies per wafer [D/W]} \times \text{Yield}}$

(2) Dies per wafer [D/W] $\approx \frac{\text{Wafer area}}{\text{Die area}}$

(3) $\text{Yield} = \frac{1}{(1 + (\text{Defect per area [Def/A]} \times \text{Die area}))^{N=2}}$

(4) $\text{Performance} \propto \frac{1}{\text{Execution time}}$

(5) CPU time [Execution time] = CPU clock cycle × Clock cycle Time [CCT]

(6) CPU Time = IC × CPI × CCT

(7) Clock cycle = IC × CPI

(8) $\text{CPU time} = \frac{\text{CPU clock cycle}}{\text{Clock Rate [f]}}$

Avg CPI

(9) $\text{Clock cycles} = \sum_{i=1}^{n} (\text{Instruction Count} \times \text{CPI})$

(10) $\text{CPI} = \frac{\text{clock cycle}}{\text{Instruction count}}$

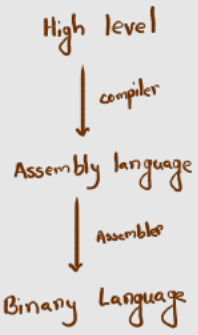
Performance [optional]:

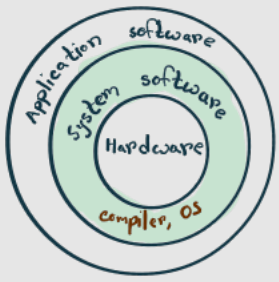
1. Algorithm
2. Language, compiler, architecture
3. Processor and memory system
4. I/O system (including OS)

Part 01

7 Great ideas in Computer Architecture:

1. Use **abstraction** to simplify design



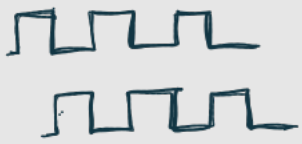


ISA (Instruction Set Architecture): set of instructions a CPU understands and executes

2. Make **common case** fast

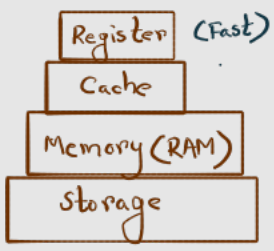
3. Perform via **parallelism**

4. Perform via **pipelining**



5. Perform via **prediction**

6. **Hierarchy** of memories





1. Volatile
2. Non volatile

7. **Dependability** by redundancy

* Inside of a processor (CPU):

- Datapath : The part of CPU that does the processing
- Control : The part that controls the data path
- Cache memory : Small but fast SRAM

History:

Vacuum tube ⟶ Transistor ⟶ IC ⟶ VLSI ⟶ Ultra large scale IC

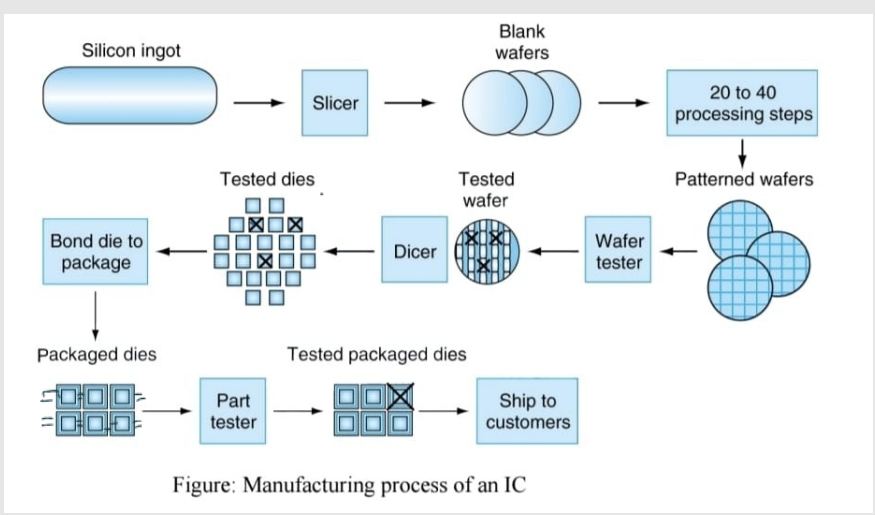
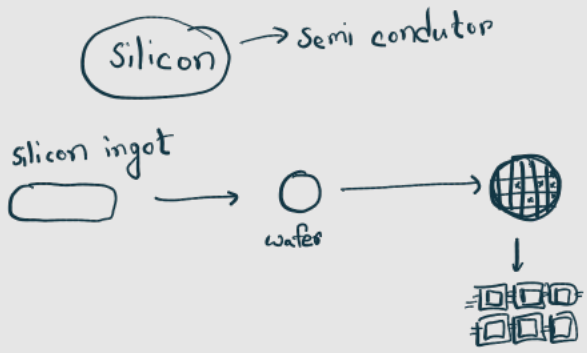


Figure: Manufacturing process of an IC



Yield : Percentage of good die from total no of die from wafer

Theory: what does 70% yield mean?

Die / Chip : Rectangular sections cut from wafer

Formula:

(1) $$\text{Cost per die [C/D]} = \frac{\text{Cost per wafer [C/W]}}{\text{Dies per wafer [D/W]} \times \text{Yield}}$$

(2) $$\text{Dies per wafer [D/W]} \approx \frac{\text{Wafer area } [\pi r^2]}{\text{Die area } [l \times r]}$$

(3) $$\text{Yield} = \frac{1}{\left(1 + (\text{Defect per area [Def/A]} \times \text{Die area})\right)^{N=2}}$$

N = 2 ↳ value যা বসবে

* Theory: Why is ≈ sign used in D/W formula?
  Ans: Chips are square but wafers are round. There is wastage due to curved edges

Assume a 19cm (diameter) wafer cost 20, contains 89 dies and has 0.023 defects/cm². Find yield.

Wafer diameter = 19cm
r = 9.5cm
C/W = 20
D/W = 89
Def/Area = 0.023

$$D/W = \frac{\pi (9.5)^2}{\text{Die area}}$$ * Careful!

$$\Rightarrow 89 = \frac{\pi (9.5)^2}{\text{Die area}}$$

$$\Rightarrow \text{Die are} = 3.18$$

$$\text{Yield} = \frac{1}{(1 + (0.023 \times 3.18))^2} = 0.87$$

Parameter for performance:

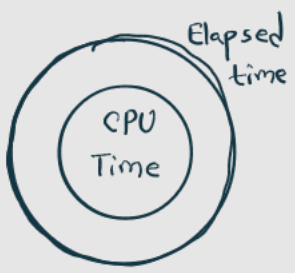
Fast processor
↳ • Response time: The time between start and completion of a task

• Throughput : Total work done per unit time

same processor but twice

• Elapsed time : The time including CPU idle time, OS overhead, I/O to complete a task
[Response/execution time]

• CPU time : The time excluding CPU idle time, OS overhead, I/O to complete a task



$$\text{Performance} = \frac{1}{\text{Execution time}}$$

Time to run a program : A = 10s , B = 15s. Who is faster?

$$n = \frac{P_A}{P_B} = \frac{E_B}{E_A}$$

$$n = \frac{15}{10}$$

$$= 1.5$$

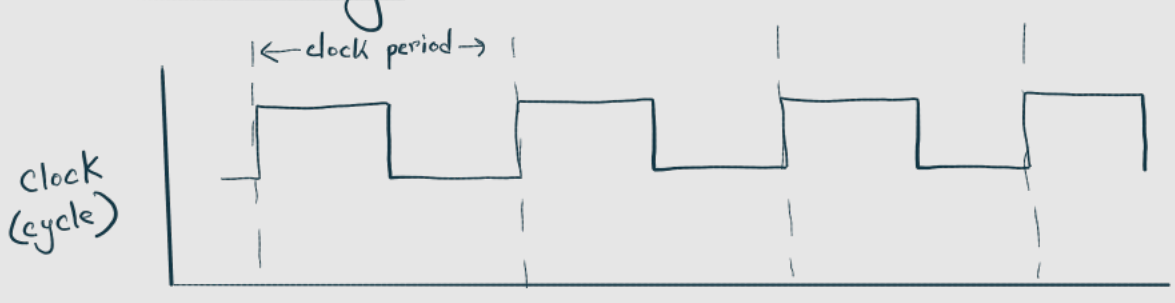
$$\frac{E_B}{E_A} = 1.5$$

$$\Rightarrow E_B = 1.5 \times E_A$$

↖ faster

Part - 2

CPU clocking :



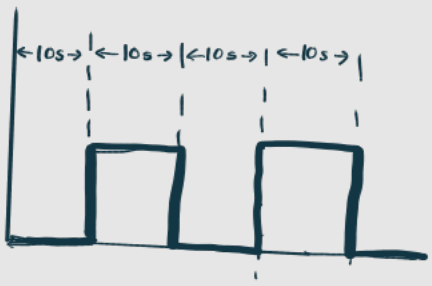
- Clock period / duration / cycle time : Time to complete one clock cycle
- Clock frequency / rate : Clock cycle completed per second. [unit: Hz]

$$f = \frac{1}{T}$$

Question Type :

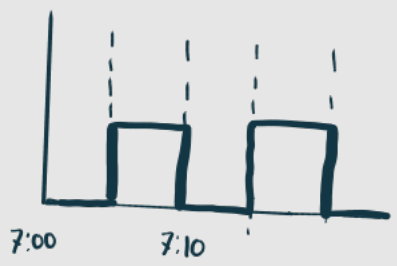
① It takes 5s to complete a clock cycle

②



clock period = 20s

③



clock period = 10s

Formula :

(1) CPU time [Execution time] = CPU clock cycle × Clock cycle Time [CCT]

= Instruction count [IC] × Cycle per instruction [CPI] × Clock cycle Time [CCT]

(2) CPU Time = IC × CPI × CCT ☆

(3) Clock cycle = IC × CPI

(4) $$\text{CPU time} = \frac{\text{CPU clock cycle}}{\text{Clock Rate [f]}} = \frac{IC \times CPI}{f}$$

* Instruction count ⟶ program, ISA (Instruction set architecture), compiler

For A,

$CPU\ time_A = 10s$

$Clock\ rate_A = 2GHz$

$Clock\ cycle_A = ?$

For B,

$CPU\ time_B = 6s$

$Clock\ cycle_B = 1.2 \times Clock\ cycle_A$

(a) Find clock cycle of A

(b) Find clock rate of B

(c) How fast the clock of computer B is?

(a) $CPU\ time = \frac{CPU\ clock\ cycle}{Clock\ Rate\ [f]}$

$\Rightarrow 10 = \frac{clock\ cycle}{2 \times 10^9}$

$\Rightarrow clock\ cycle = 20 \times 10^9$

(b) $CPU\ time_B = \frac{CPU\ clock\ cycle_B}{Clock\ Rate_B\ [f]} = \frac{1.2 \times clock\ cycle_A}{clock\ rate_B}$

$\Rightarrow 6s = \frac{1.2 \times 20 \times 10^9}{clock\ rate_B}$

$\Rightarrow clock\ rate_B = 4\ GHz$

(c) $(4 - 2) = 2\ GHz$ A is faster

It takes 15 clock cycles to run a code for CPU_A. 15s for OS overhead CPU_A has 15 GHz clock. Find the execution time

$clock\ cycle = 15$

$OS\ overhead = 15$

$f = 15 \times 10^9\ Hz$

$Execution\ time = \frac{15}{15 \times 10^9} + 15$

For the same ISA,

(A)

Cycle time = 250ps

CPI = 2.0

(B)

Cycle time = 500ps

CPI = 1.2

which is faster?

$$\frac{\text{CPU Time}_B}{\text{CPU Time}_A} = \frac{\cancel{IC} \times CPI_B \times \text{Cycle time}_B}{\cancel{IC} \times CPI_A \times \text{Cycle time}_A}$$

$$= \frac{2 \times 250 \times 10^{-12}}{1.2 \times 500 \times 10^{-12}} = 1.2$$

A faster

peta	P	10^{15}
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
hecto	h	10^{2}
deka	da	10^{1}
base unit		10^{0}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
[illegible]	[illegible]	[illegible]
pico	p	10^{-12}

Average CPI

(1) Clock cycles $= \sum_{i=1}^{n} (\text{Instruction Count} \times CPI)$

(2) $CPI = \frac{\text{clock cycle}}{\text{Instruction count}}$

$$\frac{CPI}{\sum CPI_i} = \frac{\sum IC_i}{IC}$$

↳ Relative frequency

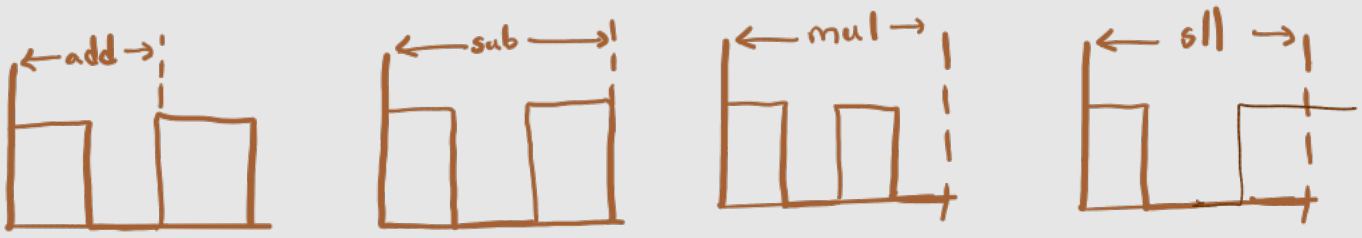
	A	B	C
CPI	1	2	3
IC	2	1	2

(a) Find clock cycle

(b) Find average CPI

(a) Clock cycle $= (2 \times 1) + (1 \times 2) + (2 \times 3) = 10$

(b) Average $CPI = \frac{10}{2+1+2} = 2$



Clock duration is 2s

Find CPU time

add ____ ✓
sub ____
add ____ ✓
sub ____
add ____ ✓
mul ____
sll ____

(7)

$$\text{CPU Time} = IC \times CPI \times C.CT$$

$$= 7 \times CPI \times 2$$

	add	sub	mul	sll
IC	3	2	1	1
CPI	1	1.5	2	1.25

$$CPI = \frac{1\times3 + 1.5\times2 + 1\times2 + 1\times1.25}{7} = \frac{9.25}{7}$$

$$\text{CPU Time} = 7 \times \frac{9.25}{7} \times 2 = \boxed{\quad}$$

Processor A has a clock cycle time of 4ns, average CPI of 0.7. P2 has a clock cycle time of 3.5ns, an average CPI of 0.7. Both of the computers have the same ISA.
Determine which computer is faster and by **how** much? [3+2]

(A)

$CCT_A = 4ns$

$Avg\ CPI_A = 0.7$

(P2)

$CCT_{P2} = 3.5ns$

$Avg\ CPI_{P2} = 0.7$

$$\frac{\text{CPU Time}_A}{\text{CPU Time}_{P2}} = \frac{\cancel{IC} \times CPI_A \times CCT_A}{\cancel{IC} \times CPI_{P2} \times CCT_{P2}}$$

$$= \frac{0.7 \times 4}{0.7 \times 3.5}$$

$$= 1.142$$

Suppose you have a brand new processor called "ProcessorX " which generates 4.5E12 instructions while executing a program. The instructions are divided into classes as follows: 10% class A, 20% class B, 50% class C, and 20% class D. It comes with a clock rate of 2.5 GHz and CPIs of 1, 2, 3, and 3 respectively for each class of instructions. **Find** the increase in CPU time if the number of instructions is increased by 10% and the average CPI is increased by 5%. [5]

(X)

$IC = 4.5\times10^{12}$; $f = 2.5\times10^{9}$ Hz

	A	B	C	D
IC	$4.5\times10^{12}\times\frac{10}{100}$	$4.5\times10^{12}\times\frac{20}{100}$	$4.5\times10^{12}\times\frac{50}{100}$	$4.5\times10^{12}\times\frac{20}{100}$
CPI	1	2	3	3

$$\text{CPU Time}_{old} = \frac{IC\times CPI}{\text{clock rate}} = \frac{4.5\times10^{12}\times \frac{\sum(IC\times CPI)}{4.5\times10^{12}}}{2.5\times10^{9}} = 4680\text{s}$$ (2.6)

Change :

$$IC_{new} = 4.5\times10^{12} + \left(\frac{10}{100}\times 4.5\times10^{12}\right)$$
$$= 4.95\times10^{12}$$

$$\text{Avg CPI} = 2.6 + \left(\frac{5}{100}\times 2.6\right) = 2.73$$

$$\text{CPU Time}_{new} = 5405.54 \quad \text{(same formula)}$$

$$\therefore \text{Increase} = (5405.54 - 4680)$$
$$= 725.4\text{s}$$

A processor executes 4×10^9 instructions: 40% ALU, 40% Memory, and 20% Control. Their CPIs are 2, 10, and 4, respectively. The clock rate is 2.5 GHz.

For $p > 1$, ALU and Memory instruction counts are divided by $(0.8 \times p)$, while Control instructions remain unchanged. For $p = 1$, use the original instruction counts.

a) Find the CPU time for $p = 1$ and $p = 4$, and calculate the relative speedup.

b) If the ALU CPI doubles from 2 to 4, find the new CPU time for $p = 4$.

c) Find the Memory CPI required for $p = 1$ to match the original CPU time for $p = 4$.

(a) $IC = 4 \times 10^9$; $f = 2.5 \times 10^9$ Hz

p = 1

	ALU	Memory	Control
IC	1.6×10^9	1.6×10^9	0.8×10^9
CPI	2	10	4

$$\text{CPU Time} = \frac{IC \times CPI}{f} = \frac{4 \times 10^9 \times \frac{\Sigma(IC \times CPI)}{4 \times 10^9}}{2.5 \times 10^9}$$

$$= 8.96$$

p = 4

$IC = 4 \times 10^9$; $f = 2.5 \times 10^9$ Hz

	ALU	Memory	Control
IC	$\frac{1.6 \times 10^9}{0.8 \times 4}$	$\frac{1.6 \times 10^9}{0.8 \times 4}$	0.8×10^9
CPI	2	10	4

$$\text{CPU Time} = \frac{IC \times CPI}{f} = \frac{4 \times 10^9 \times \frac{\Sigma(IC \times CPI)}{4 \times 10^9}}{2.5 \times 10^9}$$

$$= 2.43$$

(b) p = 4

$IC = 4 \times 10^9$; $f = 2.5 \times 10^9$ Hz

	ALU	Memory	Control
IC	$\frac{1.6 \times 10^9}{0.8 \times 4}$	$\frac{1.6 \times 10^9}{0.8 \times 4}$	0.8×10^9
CPI	4	10	4

$$\text{CPU Time} = \frac{IC \times CPI}{f} = \frac{4 \times 10^9 \times \frac{\Sigma(IC \times CPI)}{4 \times 10^9}}{2.5 \times 10^9}$$

$$= 4.08s$$

(c) $T_4 = 2.43s$

$IC = 4 \times 10^9$; $f = 2.5 \times 10^9$ Hz

P=1

	ALU	Memory	Control
IC	1.6×10^9	1.6×10^9	0.8×10^9
CPI	2	x	4

$$2.43 = \frac{(1.6 \times 10^9 \times 2) + (1.6 \times 10^9 \times x) + (0.8 \times 10^9) \times 4}{2.5}$$

$x =$ □